

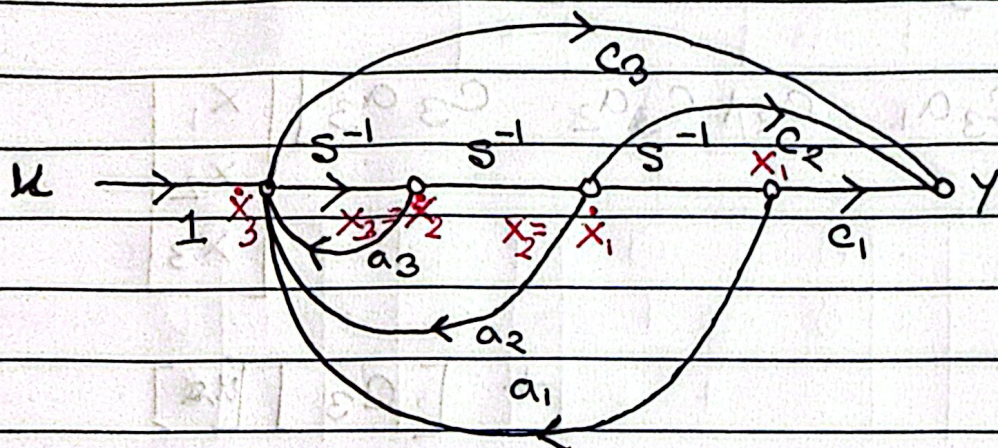
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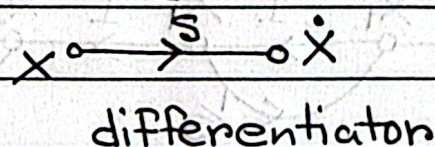
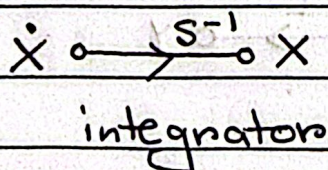
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## State Space Analysis

1. We know, state space eq<sup>n</sup>,  $\dot{X} = AX + BU$   
output eq<sup>n</sup>,  $Y = CX + DU$



Derive state model of given SFG



Total no. of integrator = Total no. of state variables

state eq<sup>n</sup>.  $\dot{x}_1 = x_2$

$\dot{x}_2 = x_3$

$\dot{x}_3 = a_1 x_1 + a_2 x_2 + a_3 x_3 + u$

|             |   |       |       |       |       |   |   |
|-------------|---|-------|-------|-------|-------|---|---|
| $\dot{x}_1$ |   | 0     | 1     | 0     | $x_1$ |   | 0 |
| $\dot{x}_2$ | = | 0     | 0     | 1     | $x_2$ | + | 0 |
| $x_3$       |   | $a_1$ | $a_2$ | $a_3$ | $x_3$ |   | 1 |

$[u]$

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output eq<sup>n</sup>:

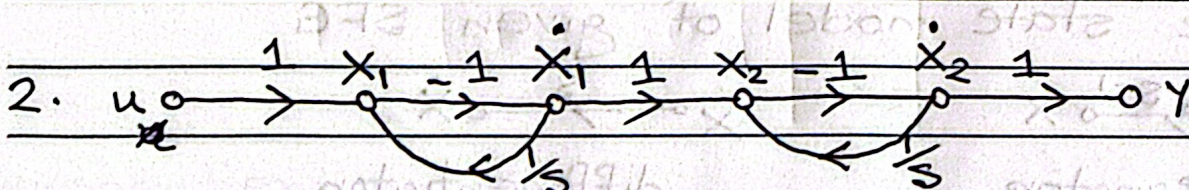
$$y = c_1 x_1 + c_2 x_2 + c_3 \dot{x}_3$$

$$= c_1 x_1 + c_2 x_2 + c_3 (a_1 x_1 + a_2 x_2 + a_3 x_3 + \cancel{u})$$

$$= (c_1 + c_3 a_1) x_1 + (c_2 + c_3 a_2) x_2 + c_3 a_3 x_3 + c_3 \cancel{u}$$

$$[Y] = \begin{bmatrix} c_1 + c_3 a_1 & c_2 + c_3 a_2 & c_3 a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$+ [c_3] [\cancel{u}]$$



Derive state model.

$$\text{State transition matrix } \Phi = L^{-1} [(sI - A)^{-1}]$$

No. of st. variables = 2

$$\text{st. eq}^n: \dot{x}_1 = -x_1 = -[u + \frac{1}{s} \dot{x}_1]$$

$$\dot{x}_1 = -[u + x_1]$$

$$\dot{x}_1 = -x_1 - u$$

$$\dot{x}_2 = -x_2 = -[\dot{x}_1 + \frac{1}{s} \dot{x}_2]$$

$$= -[-x_1 - u + x_2]$$

$$= x_1 - x_2 + u$$

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$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} [u]$$

O/p eq<sup>n</sup> :  $\dot{x} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$$[y] = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \end{bmatrix} [u]$$

Now,  $\Phi = \mathcal{L}^{-1} [(sI - A)^{-1}]$

$$\Rightarrow sI - A = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} s+1 & 0 \\ -1 & s+1 \end{bmatrix}$$

$$[sI - A]^{-1} = \frac{\text{Adj} [sI - A]}{\text{magnitude} [sI - A]}$$

$$= \frac{\begin{bmatrix} s+1 & 0 \\ 1 & s+1 \end{bmatrix}}{(s+1)^2 - (-1 \times 0)}$$

$$= \begin{bmatrix} (s+1)^{-1} & 0 \\ (s+1)^{-2} & (s+1)^{-1} \end{bmatrix}$$

Now,  $\Phi = \mathcal{L}^{-1} \begin{bmatrix} (s+1)^{-1} & 0 \\ (s+1)^{-2} & (s+1)^{-1} \end{bmatrix}$

$$= \begin{bmatrix} e^{-t} & 0 \\ t e^{-t} & e^{-t} \end{bmatrix}$$

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3. Draw an SFG for the st. space following eq<sup>n</sup> given.

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -3 & 1 \\ -3 & -4 & -5 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$$

$$Y = [1 \ 2 \ 0] X$$

Sol<sup>n</sup>:

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \dot{X}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -3 & 1 \\ -3 & -4 & -5 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$$

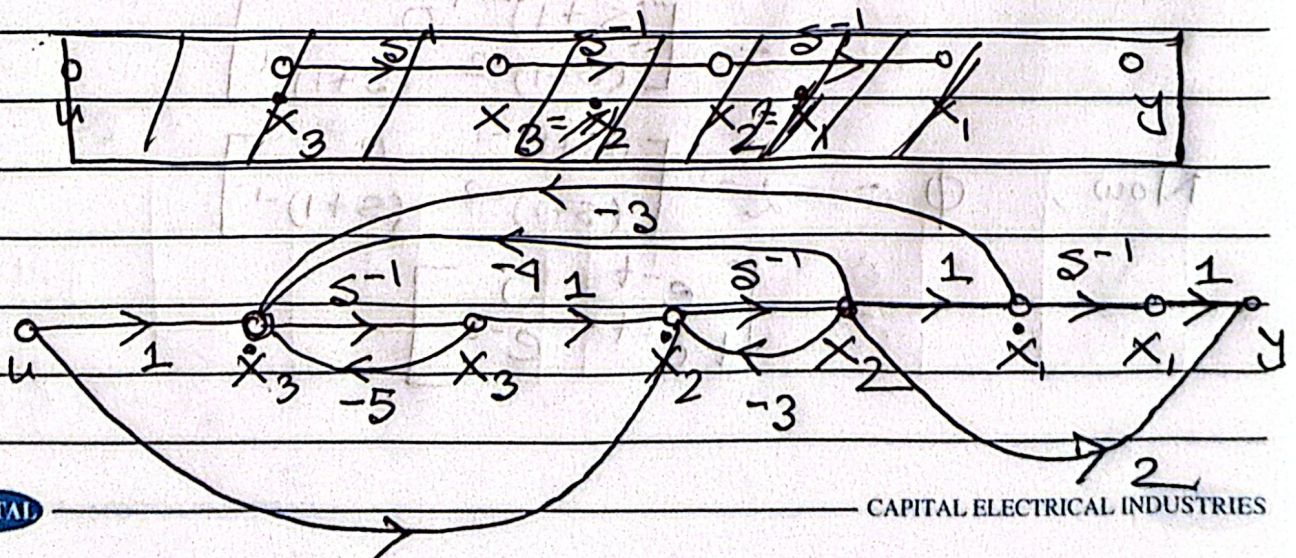
$$Y = [1 \ 2 \ 0] \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$$\dot{X}_1 = X_2$$

$$\dot{X}_2 = -3X_2 + X_3 + u$$

$$\dot{X}_3 = -3X_1 - 4X_2 - 5X_3 + u$$

$$Y = X_1 + 2X_2$$



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